

NCA Sponsored Training

Day 1 – Afternoon Session (Theory)

Cryptography 101

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Outline



1. Introduction to Cryptography

- *Goals*
- *Basic Terminology*
- *Approaches*

2. History of Cryptography

- *Classic Cyphers*

3. Symmetric Encryption

- *Modern Algorithms*



Goals of Cryptography



Problem

- Ensure **security** of communication over an **insecure** communication channel

Secure Communication

- *Confidentiality*: only the intended recipient can see the communication
- *Integrity*: the communication is generated by the alleged sender

Insecure channel

- the adversary can eavesdrop
- the adversary has full control over the communications





Approaches to Secure Communication



Steganography

- “covered writing”
- hides the existence of a message
- depends on secrecy of method

Cryptography

- “hidden writing”
- hide the meaning of a message
- depends on secrecy of a short key, not method





Goals



1. Prevent sensitive data from being read without authorization
2. Detect any unauthorized tampering of sensitive data
3. Implement non-repudiation (digital signatures)





Basic Terminology



- **Plaintext:** original message
- **Ciphertext:** coded message
- **Cipher:** algorithm for transforming plaintext to ciphertext
- **Key:** info used in cipher known only to sender/receiver



Basic Terminology



- **Encipher** (encrypt): converting plaintext to ciphertext
- **Decipher** (decrypt): recovering plaintext from ciphertext
- **Cryptography**: study of encryption principles/methods
- **Cryptanalysis** (codebreaking): study of principles/methods of deciphering ciphertext without knowing the key
- **Cryptology**: field of both cryptography and cryptanalysis



History of Cryptography



- 2500+ years
- An ongoing battle between codemakers and codebreakers
- Driven by communication & computation technology
 - *paper and ink*
 - *cryptographic engine & telegram, radio*
 - *modern cryptography: computers & digital communication*



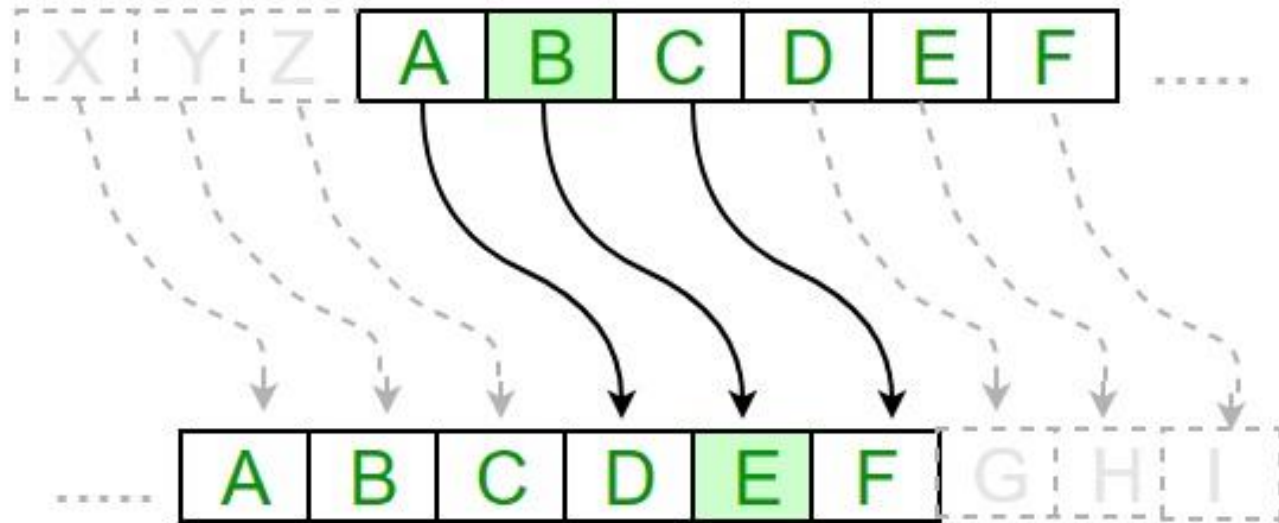
Classical Substitution Ciphers



- Letters of plaintext are replaced by other letters or by numbers or symbols

OR

- if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns





Shift Cipher



- Also known as Caesar Cipher
- Shift (Rotate) letter by agreed-upon number
- Recipient shifts back by the same number of letters
- Used for hundreds of years after Caesar's time!



A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z A B C D E F G H I J K L M N O P Q R S T U V W X Y

SAUDI = RZTCH



Shift Cipher - Cryptanalysis



Can an attacker find K?

- Only 25 possible ciphers (A maps to B, C, ... Z)
- Could simply try each in turn
- Given a ciphertext, just try all shifts of letters

KEY	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
1	oqqv	oq	chvqt	vjq	vqic	rectva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrpc	rfc	rmey	nyprw
6	jbbq	jb	xcqbo	geb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	ojbv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fqhjo
14	btti	bt	puitg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgr	gur	gbtn	cneql
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnc	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzkk	znk	zumg	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxc



Mono-alphabetic Cipher



- Rather than just shifting the alphabet, could shuffle the letters arbitrarily
- Each plaintext letter maps to a different random ciphertext letter
- The key space: all permutations of $\Sigma = \{A, B, C, \dots, Z\}$

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
B A D C Z H W Y G O Q X S V T R N M L K J I P F E U

SAUDI = LBJCG



Can an attacker find K?

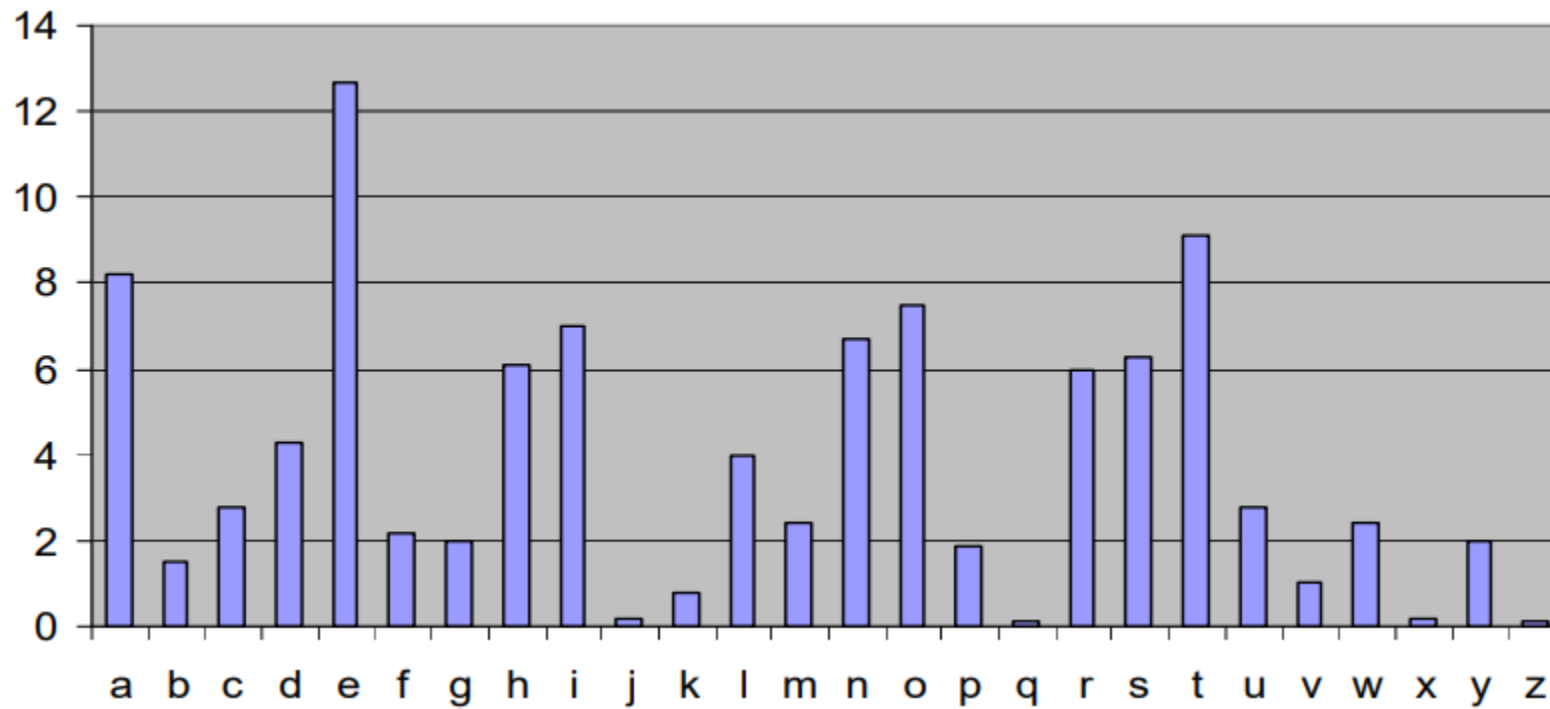
- Exhaustive search is difficult (key space size is $26! \sim 4 \times 10^{26}$)
- Dominates the art of secret writing throughout the first millennium A.D.
- Thought to be unbreakable by many back then



Substitution Ciphers - Cryptanalysis



Frequency of Letters in English





Substitution Ciphers - Cryptanalysis



Frequency Analysis

- Each language has certain features: frequency of letters, or of groups of two or more letters.
- Substitution ciphers preserve the language features.

By analyzing letter frequencies and comparing them to standard frequencies in the language used, substitution ciphers are **easily cracked**. Al- Kindi, 870





One-time Pad



- Fix the vulnerability of the mono-alphabetical
- substitution cipher by encrypting letters in different locations differently
- Key is a random string that is at least as long as the plaintext
- Encryption is similar to shift cipher
- Invented by Vernam in the 1920s



One-time Pad



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

E 4 + K 10 = 14

N 13 + E 4 = 17

I 8 + Y 24 = 6

G 6 + W 22 = 2

M 12 + O 14 = 0

A 0 + R 17 = 17

plain text: ENIGMA

keyword: KEYWORD

ciphertext: ORGCAR



Security of One-time Pad



- OTP has unconditional security
 - *OTP is information-theoretically secure (or provably secure)*
- Ciphertext C gives absolutely no additional information about the plaintext

Example: Eve intercepts Alice's ciphertext : "EQNVZ"

Eve can find that key "XMCKL" would produce plaintext "HELLO", but she would also find that key "TQURI" would produce plaintext "LATER", equally plausible message



- **Practical limitations**

- *Making large quantities of random keys*
- *Problem of key distribution and protection, where for every message to be sent, a key of equal length is needed by both sender and receiver.*

- **Use of one-time pads**

- *British Special Operations Executive used OTPs in World War II*
- *Hotline between Moscow and Washington DC (established after the Cuban missile crisis) used a commercial one-time tape system*



Security Principles



- Security by obscurity does not work!
- The security of the system cannot rely on keeping the algorithm(s) secret.
- Should assume that the adversary knows the algorithm; the only secret the adversary is assumed to not know is the key



Confidentiality:

- Ensuring information is disclosed only to those authorized to know it.

Threats to consider:

- Data in transit (packet sniffing/wiretap/surveillance, Man-in-the-middle, etc.)
- Data at rest (DAR): (file grabbing from a compromised system, stolen laptop, etc.)



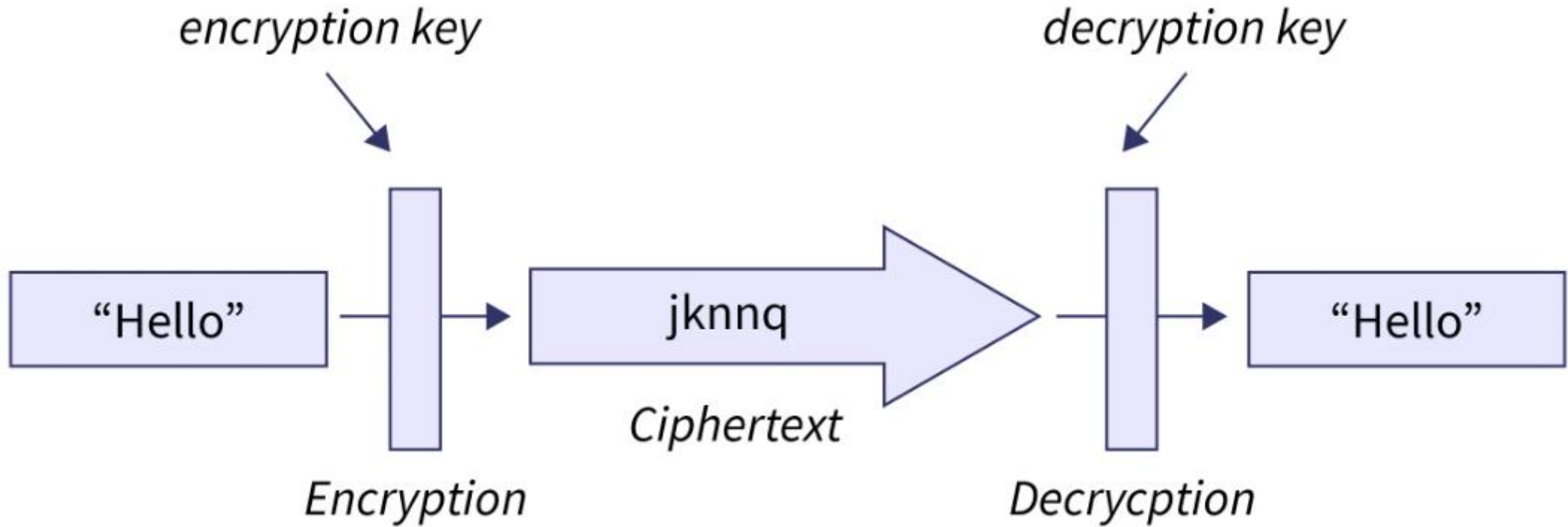
What Can We Enforce? PAIN



- **Privacy:** Transactions cannot be viewed by an unauthorized party.
- **Authentication:** Assurance that an entity's identity claim is valid.
- **Integrity:** Assurance that data has not been altered either intentionally or unintentionally.
- **Non-repudiation:** Incontrovertible evidence of what occurred and who was involved.

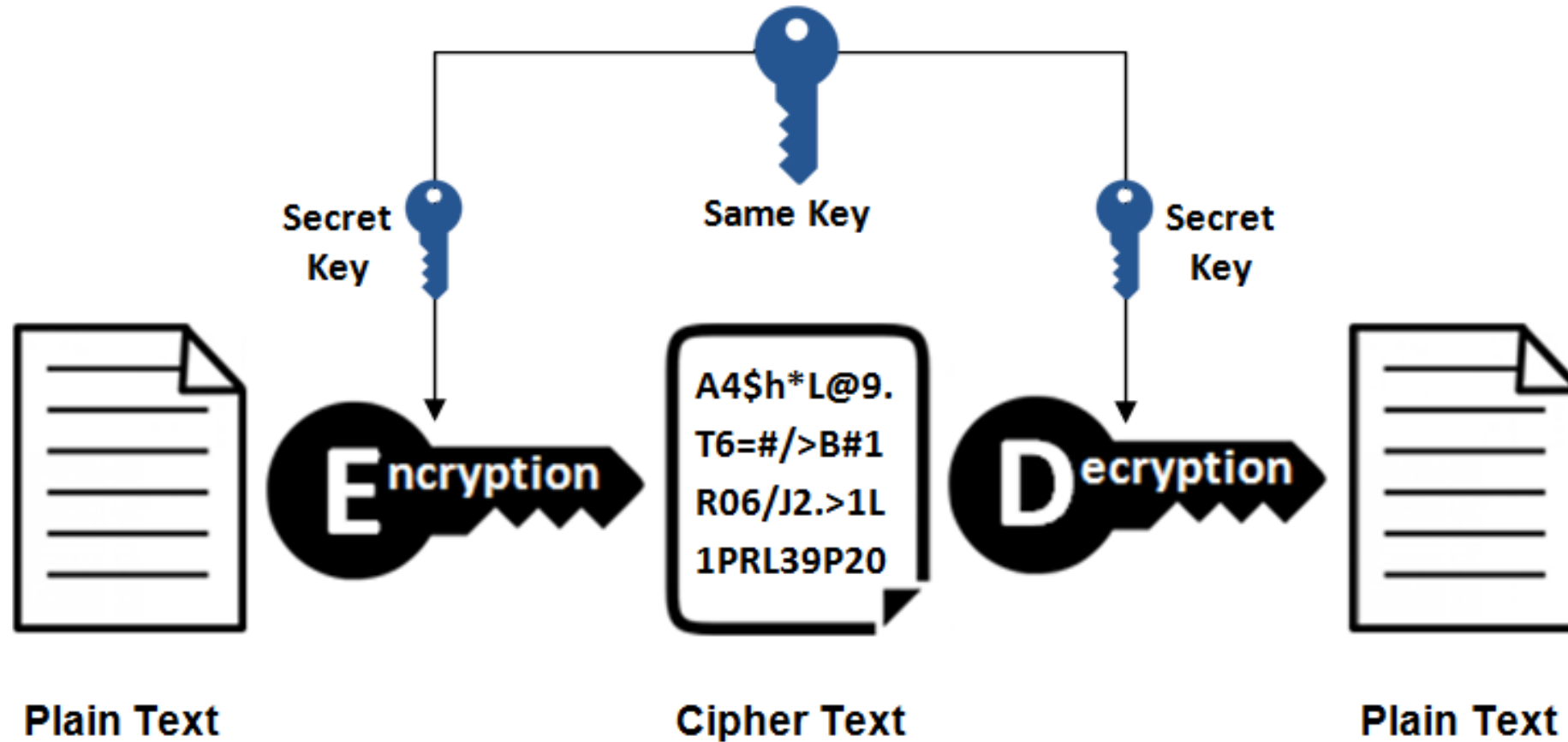


Cryptography Process





Symmetric Cryptography





Block ciphers process messages in blocks, each of which is then en/decrypted (like a substitution on very big characters, e.g., 64 bits)

Why Block Ciphers?

- Another way to defeat frequency analysis
- Make the unit of transformation larger (rather than encrypting letter by letter, encrypting block by block)



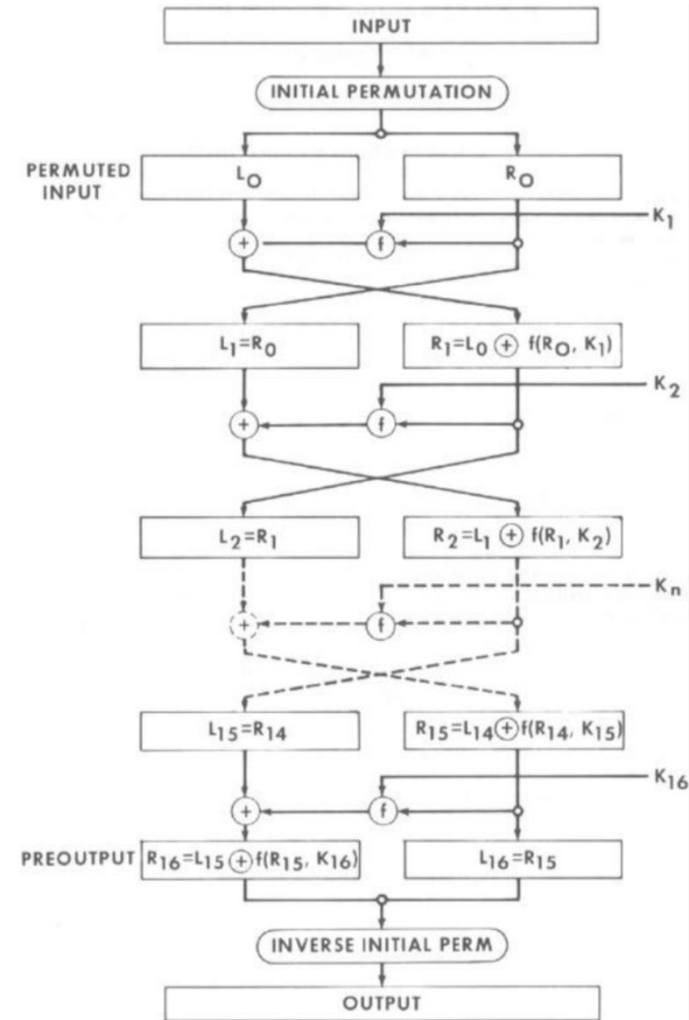
- Data Encryption Standard (DES)
 - *Designed by IBM with modifications proposed by the National Security Agency*
 - *US national standard from 1977 to 2001*
- Advanced Encryption Standard (AES)
 - *Originally developed by two Belgian cryptographers*
 - *Became standard for US government in 2001*



Data Encryption Standard (DES)



- Symmetric-Key block cipher algorithm
- Encrypts 64-bit blocks using a 56-bit key
- 16 iterations: replacing blocks of bits (**substitution**), shuffling the bits (**permutation**), mingling in bits from the key (**transformation**)
- Designed mostly for hardware implementations





Data Encryption Standard (DES)



- Broken in 1999: **56-bit key** is too **small** (vulnerable to brute-force attacks)
- NIST **withdrew** DES as an encryption standard in 2005

Other versions:

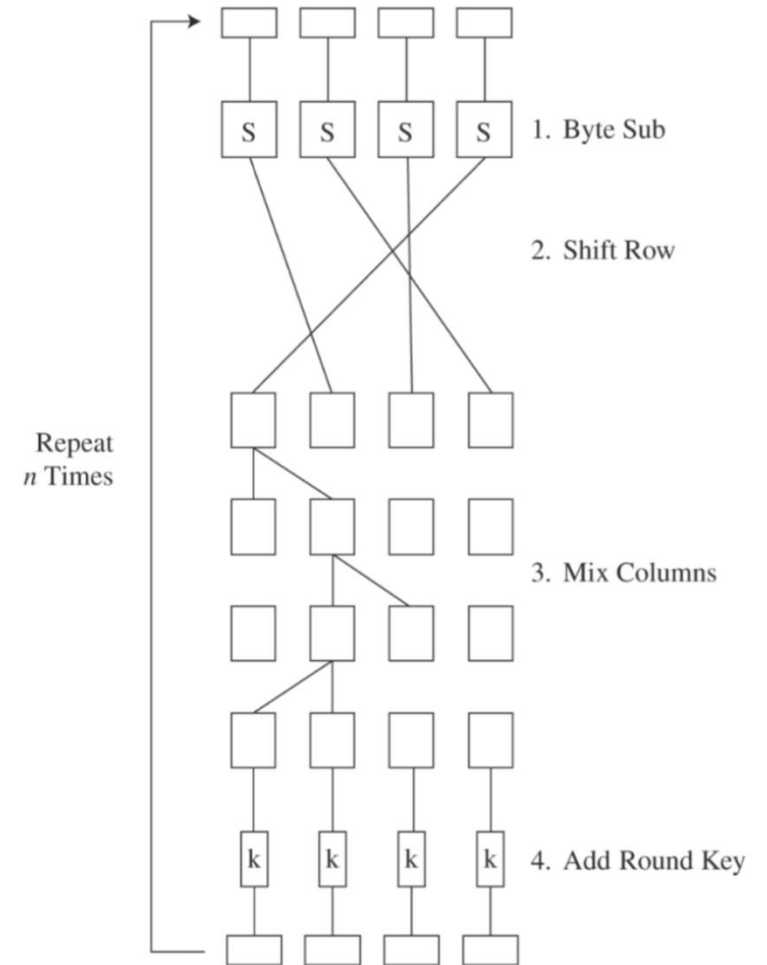
- **Double DES:** $C = E(K2, E(K1, M))$ just doubles the length of time to break this version
- **Triple DES:** $C = E(K3, E(K2, E(K1, M)))$ Encryption strength up to 112-bit key



Advanced Encryption Standard (AES)



- Goal: replace DES for both government and private-sector encryption.
- Designed to be efficient in both hardware and software across a variety of platforms.
- Block size: 128 bits
- Variable key size: 128, 192, or 256 bits.
- Variable number of rounds (10, 12, 14) depending on the Key size





Symmetric Key Summary



- Both sides use the same key to encrypt & decrypt
- Advantages
 - *Fast – suitable for real-time and bulk encryption of high-speed communication systems*
- Disadvantages
 - *Key Distribution*
 - *Key Management*



Cryptography VS Steganography



- Cryptography is about **protecting** the content of messages (their meaning).
- Steganography is about **concealing** the existence of messages



Steganography - Basics



Steganography simply takes one piece of information and hides it within another:

- *Computer files (images, sounds recordings, etc.) contain unused or insignificant areas of data*
- *Steganography takes advantage of these areas, replacing them with information.*
- *The files can then be exchanged without anyone knowing what really lies inside of them*



Steganography – Early Examples



In his history of the Persian Wars, Herodotus tells of a messenger who shaved his head and allowed a secret message to be tattooed on his scalp. He waited until his hair grew back. Then he journeyed to where the recipient awaited him and shaved his head again. The message was revealed. It was history's first use of steganography.





Steganography – Other Examples



- **Roman Empire:** Ancient Romans used to write between lines using invisible ink based on various natural substances.
- **World War II:** A secret message was photographically reduced to the size of a period, and affixed as the dot for the letter 'i' or other punctuation on a paper containing a written message.



Computer Steganography



Computer Steganography is based on two principles.

1. Files containing digitized images or sound can be altered to a certain extent without losing their functionality.
2. The other principle deals with the human inability to distinguish minor changes in image color or sound quality, which is especially easy to make use of in objects that contain redundant information. The value of the least significant bit of the pixel color won't result in any perceivable change of that color.



Steganography - Process



- The data to be concealed is compressed and hidden within another file.
- The first step is to find a file which will be used to hide the message (also called a carrier or a container.)
- The next step is to embed the message one wants to hide within the carrier using a steganographic technique.

Most used embedding technique: replace the least significant bit of each byte in the [carrier] with a single bit for the hidden message.



Steganalysis



The art of discovering and rendering useless such covert messages.

- Steganalysis involves analysis of the carrier file
- Simpler steganographic techniques produce some noticeable change in the file size, statistics or both.
- These changes can manifest themselves in color variations, loss of resolution and other distortions that are visible to the human eye.
- This form of detection requires that you know what the original carrier image or file should look like.

Do you have any questions?



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